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The role of populist attitudes in explaining climate change skepticism and support for environmental protection

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ABSTRACT
Conventional wisdom holds that partisanship and political ideology, writ large, are some of the most powerful explanations of attitudes towards climate change and environmental politics. While compelling, most studies focus on a narrow definition of political ideology in the US. This study adds to the literature by assessing the relationship between populism, climate skepticism, and support for environmental protection. Populism offers an orthogonal dimension to partisanship and left-right self-placement, which broadens the scope of the concept. Assessing the UK facilitates understanding the role of political ideology beyond the strong party sorting apparent in the US. Data from the 2015 British Election Study offer strong support for the proposition that populism holds a consequential role in climate and environmental politics.

KEYWORDS Populism; political ideology; climate skepticism; environmental politics; climate change

Introduction

Understanding citizens' perceptions of climate and environmental politics is essential. Depending on the circumstances, citizens pressure governments to act on climate change and environmental degradation, or in turn, governments may seek to induce pro-environmental behaviour through policies. Thus, governments have adopted international accords, such as the 2015 Paris Agreement – building on scientific evidence highlighting anthropogenic causes of climate change – and actively communicated reasons for taking such action. While many politicians and scholars concur on the necessity of taking mitigating actions, not all citizens share this sense of urgency. In fact, many British citizens disagree with the statement that the earth is warming due to human activities (YouGov 2014, Fieldhouse *et al.* 2015), while others equate environmental tendencies to elitism (Morrison and Dunlap 1986, Wetts 2019). In 2012, YouGov reported that only

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Supplemental data for this article can be accessed here.

20 per cent of UK citizens entirely attribute climate change to humans, while 15 per cent deny any influence of humans. 61 per cent do not attribute sole responsibility to humans or do not know (5 per cent). Capstick and colleagues (2015) confirm similar figures. Importantly, even if an individual acknowledges climate change, denying the human induced component undermines willingness to act, since it is then out of individuals' hands. Consequently, given that public support is an essential condition for far-reaching climate and environmental policies (Anderson *et al.* 2017), citizens' climate skepticism has the potential to limit progress. *But where does this skepticism come from?*

Conventional wisdom holds that partisanship and political ideology are, writ large, some of the most powerful explanations of attitudes towards climate change and environmental politics (Hornsey *et al.* 2016, Beiser-McGrath and Huber 2018). Clearly, political ideology matters even in issues such as environmental preferences. While compelling, the literature on partisan sorting suffers from two limitations. First, scholarship so far is predominantly US-centric. While the case of the United States is compelling, the dynamics of political ideology and partisanship make it unique. Second, most have adopted a narrow conceptualisation of political ideology. So far, the assessment of the role of political ideology mainly focuses on left-right self-placement and party proximity, which are simplified proxies for the multifaceted concept of political ideology. Here, I address these limitations by empirically assessing the political behaviour of UK citizens, and by introducing an additional dimension of political ideology that has been overlooked so far: populism.

Evaluating the role of political ideology in the UK offers a clear advantage: here, attitudes to climate change are not as polarised along party lines as in the US. While strong party-sorting in the US obscures the differentiation of the effects of political ideologies and partisan cues (Layman *et al.* 2006, Fiorina *et al.* 2008, Guber 2013), the lack of strong party cues and populist vote in the UK allows us to home-in on the isolated demand-driven relationship between individual citizens' populist attitudes and their preferences towards climate and environmental policies.

Introducing populism into our understanding of climate skepticism is especially important. Climate change, and to some extent environmental degradation, present ideal targets for the populist backlash against globalisation (Norris and Inglehart 2018); climate change's abstract and complex nature allows populists to diminish these issues as elite projects. Until now, research linking populism and climate and environmental politics has been limited to specific circumstances, and remains theoretical (Lockwood 2018); whether populist-oriented individuals oppose climate-related and environmental politics remains empirically unclear and untested. Drawing on the literature on elite- and source-opposing cues, I argue that individuals who exhibit strong populist attitudes feel under-represented in both climate and environmental politics. When individuals perceive climate change issues as elite-driven concepts detached from their everyday needs, climate-related and environmental politics are eventually rejected. Populist attitudes, climate skepticism, and policy support for climate and environmental policy are thus inter-dependent.

Data from the 2016 pre-local election wave of the 2014–2018 British Election Study (wave 7 – Fieldhouse *et al.* 2015) provide empirical evidence that climate change perceptions are affected by rising populist sentiments. Individuals who exhibit strong populist attitudes are less likely to believe in human-induced climate change, and are more likely to oppose environmental protection. These findings are robust to alternative specifications and hold for individuals on both left and right of the political spectrums. In other words, the data suggest that populist attitudes in the UK provide an orthogonal dimension to the classic left-right spectrum.

Altogether, populism's primary focus on the central actors in policymaking – political elites – offers a novel dimension that is often ignored by explanations of climate skepticism and stances on environmental protection. Broadening the concept of political ideology and exploring so far neglected facets further enhances our understanding of climate and environmental attitudes. Especially in cases where partisanship, left-right self-placement, and populism might be conflated by party-side supply, such as the US since the candidacy of Donald Trump (Oliver *et al.* 2016), the effect of partisanship may be over-estimated. Considering populism in climate-related and environmental politics is of utmost importance to disentangle the different forces at work. The findings presented here also carry implications for the literature on the communication of scientific evidence, as well as on climate change and environmental degradation. If climate skeptics also hold anti-elitist views, this may explain potential backlashes to top-down climate communication beyond beliefs on policies (Huber *et al.* 2019). In light of my findings, politicians and scholars might want to experiment on the development of new ways to communicate the importance of climate action.

Populism, climate skepticism, and support for environmental protection

Here, I consider populism as a set of ideas exhibiting the following basic features: anti-elite attitudes; emphasis on the central role of the 'people'; and the perception that two homogeneous societal groups (the people and the elite) are caught up in a moral struggle (Mudde 2004, Rooduijn 2014, Hawkins and Rovira Kaltwasser 2018). Like populism itself, individuals may also exhibit these three core concepts, and present strong populist attitudes as a result (see, e.g. Akkerman *et al.* 2014, Hawkins and Rovira

Kaltwasser 2018). In other words, these attitudes function as the demand side of populist politics, where individuals perceive a lack of representation by the governing political elite (Hawkins and Rovira Kaltwasser 2018).

The dependent variables in this study focus on climate skepticism and its corresponding support/opposition towards environmental politics. I consider climate skepticism as a disbelief or uncertainty concerning (anthropogenic) global warming that espouses a lack of acceptance or awareness of the seriousness of climate change and its consequences (Stoll-Kleemann *et al.* 2001, p. 112), e.g. by disbelieving the anthropogenic nature of climate change (Stoll-Kleemann *et al.* 2001, Hobson and Niemeyer 2013). Thus, skepticism about environmental protection generates doubt regarding the gravity of climate change, the potential consequences of environmental degradation, and the necessity of environmental action (Jacques *et al.* 2008).

Political correlates of climate and environmental attitudes

The need to explain climate skepticism and support for environmental protection has sparked a large, and still blooming, literature. Explanations related to political ideology and other political factors, such as trust in government, are of most interest to political scientists.

The state-of-the-art suggests that political ideology explains a significant amount of climate skepticism and support for environmental protection (Dunlap *et al.* 2001, 2016, Neumayer 2004, McCright and Dunlap 2011, Guber 2013, McCright *et al.* 2016). By and large, these scholars argue that right-wing ideology is associated with lower support for environmental and climate-related regulations; right-wing voters are more likely to reject climate and environmental policies. By contrast, left-wing ideology favours state intervention and supports the regulation of behaviour that affects the climate and environment. In the US context, these studies pit conservatives and Republicans against liberals and Democrats, whereby climate and environmental attitudes are polarised and sorted along party lines (Layman *et al.* 2006, Guber 2013, Currie and Choma 2018, Horne and Huddart Kennedy 2019), hence affecting attitudes towards climate change and environmental degradation directly and indirectly. Individuals interpret information on climate change and environmental degradation in line with their worldview (Zhou 2016). For instance, Hamilton (2011) finds that the effect of education on climate concern is moderated by partisanship. While Democrats' climate concern increases with additional education, Republicans' climate concern is not associated with education. Similarly, the interpretation of personal experience with extreme weather events is conditional on partisanship (Borick and Rabe 2010). The reasoning and processing of climate-relevant information happen along party lines (Malka *et al.* 2009, Hart and Nisbet 2012).

At the same time, Pechar *et al.* (2018) show that trust in government affects trust in (climate) science (Lorenzoni and Pidgeon 2006, Gauchat 2012). Furthermore, mistrust in science decreases concern about climate change (Kellstedt *et al.* 2008) and climate policy support (Konisky *et al.* 2008), and it is easily derailed by conspiracy theories (Lewandowsky *et al.* 2013) and scandals such as 'climategate' (Leiserowitz *et al.* 2013). Finally, several authors suggest that authoritarianism (vis-à-vis liberal positions) is associated with climate skepticism (Leiserowitz 2005, Poortinga *et al.* 2011, Whitmarsh 2011).

The missing dimension: populism

Given the importance of political variables as explanations of attitudes towards climate-related and environmental politics, it is surprising that most of the literature relies on narrow conceptualisations of political ideology. While these are valid and fruitful conceptualisations, they overlook important dimensions of the political debate, namely political elites, the central actor in climate action and environmental protection. Recent events such as Brexit, Donald Trump's election, and other successes of populist actors would suggest that we should consider a broader conceptualisation since they were not (entirely) driven by left- or right-wing sentiments; instead, populist sentiments played an important role (see, e.g. Hobolt 2016, Oliver *et al.* 2016). Anti-politics and a lack of trust in the political 'establishment' (Hay and Stoker 2009), an essential subdimension of populism, creates a new dimension of political ideology that directly confronts the central actor at the heart of policymaking. Populism appears on both the political left and right and is theoretically orthogonal to the left-right spectrum (Rooduijn and Akkerman 2017).

Lockwood (2018) provides an excellent first theoretical assessment of the role of right-wing populism and climate skepticism: explanations of populists' opposition to climate change policies might adopt a structuralist or an ideological approach. The former argues that populists appeal most to lower skilled males most affected by globalisation. Climate policy directly erodes job security in low skilled manufacturing jobs most directly targeted by far-reaching regulations. The second approach is closer to the theoretical argument I pursue below. Lockwood argues that right-wing populists are socially conservative and hold strong nationalist values. As a consequence, because climate policies threaten national sovereignty, right wing-populists resist them (Lockwood 2018).

Although these arguments are plausible in and by themselves, Lockwood combines populism with nationalism and authoritarianism, in line with Mudde's definition of populist radical right parties (Mudde 2007). By design, Lockwood's study does not disentangle populism from the political ideology it attaches to. It is necessary to investigate populism across different political positions to assess the explanatory power of populism vis-á-vis other components of political ideology, such as a left-right placement. Indeed, recent studies have started to assess how ideology and populism interact, and in turn affect parties' and individuals' attitudes and behaviour (Otjes and Louwerse 2015, Akkerman *et al.* 2017, Huber and Ruth 2017, Huber and Schimpf 2017). Here, the core difference between left- and right-wing populist actors concerns their conceptualisation of 'the pure people' and 'the corrupt elite'. This can be illustrated by describing their reference in their narrative to a utopian heartland, which represents an idealised world before corruption by the elite (Taggart 2002, pp. 67–68). Populist parties of different ideological character construct different heartlands; right-wing populist parties include a strong cultural and nativist aspect (Forchtner and Kølvraa 2015, p. 199). This element is entirely lacking in the discourse of left-wing populist parties, which tend to define their heartland economically.

My argument thus builds on Lockwood's important contribution in two ways. First, I subject the arguments surrounding populism and climate attitudes to rigorous empirical tests on the individual level. Second, the design allows for the differentiation of the role of populism from other measures of political ideology, such as left-right self-placement, party proximity, and/or authoritarianism. My approach is embedded in the broader literature on how political elites can affect citizens' attitudes (Zaller 1992, Leeper and Slothuus 2014). The literature argues that individuals will use information about the sender of a message to infer information about the proposal. For example, messages from senders perceived to be untrustworthy receive little attention, or result in an unfavourable message evaluation (Aaroe 2012, Nicholson 2012). Eventually, the perception of the source affects whether individuals agree or disagree with the conclusion of a statement (McGuire 1969).

I argue this is precisely what happens with populists and attitudes towards climate change and environmental degradation. The nature of climate and environmental politics is abstract and technical, and thus populists can easily portray them as elite-driven and detached from citizens' everyday needs since the topic itself is elite-driven, of interest to richer and better-educated citizens, and is a prime example of post-materialist issues (Morrison and Dunlap 1986, Freudenberg and Steinsapir 1991, Inglehart 1995, Wetts 2019). International climate policies are primarily discussed in international fora such as the UNFCCC and the associated Conference of the Parties (COP), where the public is overtly excluded from decision-making. These general characteristics make environmental issues ideal targets for populists, who can easily perceive climate policy to be part of an elite-driven, cosmopolitan agenda (Lockwood 2018) that has lost touch with citizens' everyday needs and preferences.

The elite-driven top-down discourse in its current form will likely face difficulties in seeking to convince populist individuals to accept unpleasant alterations to daily life. The psychological distance of these issues increases due to the international nature, temporal vagueness, and uncertainty of both issues (Spence *et al.* 2012, Weber 2016). Uncertainty, in particular, ties in with populists' bias towards conspiracy theories to portray a conspiracy of elites (Castanho Silva *et al.* 2017). Moreover, since conspiracy theories are often linked with climate change denial (Lewandowsky *et al.* 2013, Hornsey *et al.* 2018), it is plausible to anticipate that populists might oppose climate-related and environmental politics.

As climate science is uncertain, climate change skeptics exploit its uncertainty to criticise climate action. For example, Sussman argues that global warming is an elite project that provides the 'elite' the opportunity to carry out individual agendas. He accuses the political and intellectual elite of using their policy position for personal gain, allowing for the 'biggest scam in history' (Sussman 2010, p. 215). Nevertheless, given the overwhelming scientific consensus regarding humans' impact on climate change, Sussman's argument is a clear example of how climate and environmental politics is often entangled with conspiracy theories (Lewandowsky *et al.* 2013).

As long as populists portray combating climate change and environmental degradation as an elite project, populist attitudes could be associated with climate skepticism and dismissal of environmental protections. Theories on elite cues and their reception help understand this relationship (Johnson *et al.* 2005, Carmichael and Brulle 2017). Cue-taking describes the adoption of positions signalled by the sender, in this case, the political elite (Steenbergen *et al.* 2007). However, there are limits to this effect: Druckman (2001) argues that source credibility is an important moderator. If individuals oppose the source of a message, they are more likely to distrust the message that is introduced or potentially even take a more negative stance on that issue (Zaller 1992, Druckman 2001, Aaroe 2012, Nicholson 2012).

While Aaroe (2012) and Nicholson (2012) specifically focus on proximity to political parties, I argue that this effect is not limited to party identification but instead reflects more general attitudes towards the political elite. If individuals distrust the political elite and perceive a moral struggle between 'the people' and 'the elite' – in other words, they strongly exhibit populist attitudes – they are more likely to reject a seemingly united elite position and take more negative stances on the relevant issue.

As communication on climate change and environmental protection is heavily driven by top-down elite cues, the nature of this policy field could result in rejection by populists. Accordingly, I hypothesise that high levels of populist attitudes are associated with skepticism about climate change and lower support for environmental policies:

- H1a: Individuals who exhibit strong populist attitudes are more likely to be skeptical about climate change.
- H1b: Individuals who exhibit strong populist attitudes are less likely to support environmental protection.

Research design

In order to assess the association between populist attitudes, climate skepticism, and stances on environmental protection, data from the seventh wave of the British Election Study (BES) are used (Fieldhouse *et al.* 2015). This survey provides the best possible combination of high-quality sampling in combination with state-of-the-art measures of populist attitudes. The measurement of populist attitudes is explicitly designed to capture individual manifestations of populism without conflating it with political ideology or other related variables such as political trust (Akkerman *et al.* 2014). Disentangling the correlation between populism and climate and environmental attitudes from political ideology and other control variables such as political trust is of utmost importance. Several empirical studies confirm the high quality of this measure in different settings and contexts (Castanho Silva *et al.* 2019, Van Hauwaert *et al.* 2019). The quality of the online survey in the British Election Study is particularly high; approximately 4,100 respondents answered questions on populism, climate skepticism, and environmental protection between 14 April and 4 May 2016. The survey included individuals older than 16 who were eligible to vote in any UK election. YouGov selected the respondents from their online panel using quotas and weights to approximate the general population.

The UK is often portrayed as harbouring substantial climate skepticism compared with the rest of Western Europe (Devine-Wright 2005, Whitmarsh 2011, Capstick and Pidgeon 2014); for example, compared to Germany, where almost no climate skepticism exists (Engels *et al.* 2013), UK citizens are noticeably more uncertain about climate change and its causes (Poortinga *et al.* 2011). Similarly, the number of climate-skeptical newspaper reports is higher in the UK than in France (Painter and Ashe 2012). Hence, the UK provides interesting variation in the outcome measures, attitudes towards climate change, and environmental protection (see Table 1). Equally, the UK has no experience of a strong populist party in the House of Commons; UKIP, an anti-EU populist party, has only ever held one seat in the national parliament, and has rarely politicised climate and environmental issues. The relationship between populist attitudes and attitudes towards climate and environment should therefore be largely driven by citizens' attitudes (demand) rather than party cues on policy positions (supply). The case of the UK thus allows us to disentangle demand- and supply-driven climate skepticism as, in contrast to the US, climate and environmental attitudes are neither polarised nor sorted along party lines (Layman *et al.* 2006, Fiorina *et al.* 2008).

Table 1. Perceived policy consequences and policy support.

Dependent Variable	Level	N	Percent
Climate Skepticism	[Climate is] changing due to human activity	4237	64.17%
	… changing but not due to human activity	1561	23.64%
	… not changing	277	4.20%
	Don't know	528	8.00%
Environmental Protection	Measures have gone too far	1169	17.70%
	Measures were about right	2033	30.79%
	Measures have gone not far enough	2990	45.28%
	Don't know	411	6.22%
Environmental Protection vis-á-vis Economic Growth	0	511	8.14%
	1	228	3.63%
	2	530	8.44%
	3	648	10.32%
	4	710	11.30%
	5	1352	21.53%
	6	571	9.09%
	7	591	9.41%
	8	463	7.37%
	9	261	4.16%
	10	416	6.62%

Dependent variables

The following analyses draw on one item to measure climate skepticism and two different measures to quantify support for environmental protection (see Table 1). First, individuals were asked whether they think that climate change is: anthropogenic; existent but not human-made; non-existent; or 'Don't know'. This question operationalises one of the key dimensions of climate skepticism: disbelief in anthropogenic climate change.[1] Second, individuals responded to the question asking whether they think government actions to protect the environment have gone too far, are about right, or have not gone far enough. Finally, participants were asked how they would balance the trade-off between environmental protection and economic growth. Respondents, on an eleven-point scale, state whether they support economic growth at the cost of environmental protection, or support environmental protection at the cost of economic growth.[2] Higher values indicate less support for environmental protection.

Independent variable

Measuring populist attitudes follows Akkerman *et al.* (2014) and includes five items. These items measure agreement and disagreement with several questions that tap into the key dimensions of populism: people-centrism, anti-elitism, and the general will of the people. An exploratory factor analysis (EFA) reveals that the items load around a single factor and are part of the same latent construct,

Table 2. Factor analysis: populist attitudes.

Question	Loading	Akkerman *et al.* (2014)
The politicians in the UK Parliament need to follow the will of the people	0.61	Pop 1
The people, and not politicians, should make our most important policy decisions	0.79	Pop 2
I would rather be represented by a citizen than by a specialized politician	0.74	Pop 4
Elected officials talk too much and take too little action	0.62	Pop 5
What people call 'compromise' in politics is really just selling out on one's principles	0.57	Pop 6

Cronbach's alpha of the six items equals 0.8. Pattern (loading) matrix from exploratory factor analysis extracting one factor from five items measuring populist attitudes. I used the varimax method to extract these factors. (Dis-)Agreement with each item was reported on a five-point scale ranging from strongly disagree (1) to strongly agree (5).

which will be used as the main independent variable in subsequent analysis (see Table 2 for the wording of each item and the factor analysis results).

Control variables

In order to estimate the association between populist attitudes and stances on climate change and environmental protection, the inclusion of control variables is important. I include party identification and political ideology, which are regularly used to explain stances towards climate and environmental politics (Dunlap *et al.* 2001). Individual characteristics and resources tend to be associated with attitudes towards climate and environmental politics. The literature suggests that younger, better-educated women with higher incomes are less skeptical about climate change and environmental degradation (Blocker and Eckberg 1997, Dietz *et al.* 1998, Whitmarsh 2011, Franzen and Vogl 2013). I enter age, gender, education (university degree or not) and income as control variables in my regression model.

Similarly, political variables such as interest in politics (on an 11-point scale from 0 'pay no attention [to politics]' to 10 'Pay a great deal of attention'), satisfaction with democracy (measured on a 4-point scale, higher values represent more satisfaction), and political efficacy (factor based on four items) have been shown to be related to both climate attitudes and populism (Pechar *et al.* 2018), as well as individual and societal economic development in the previous year (both captured on a five-point scale, with higher levels indicating better evaluations of last 12 months, Scruggs and Benegal 2012, Kachi *et al.* 2015), risk aversion (four-point scale, higher levels indicate high willingness to take risk), trust in society (three categories: 'Most people can be trusted', 'Can't be too careful' and 'Don't know'), and authoritarianism (factor based on four items) could be associated with both populist attitudes and beliefs about climate change and environmental degradation (Leiserowitz 2005, Kellstedt *et al.* 2008, Kriesi 2010, Poortinga *et al.* 2011). All these are included as control variables. Lastly, I control for partisanship to account for potential sorting along party-cues. Table A1 contains descriptive statistics for all variables; consult the corresponding section in the Appendix for the detailed wording of all used variables.

Model specifications

The measurements of climate skepticism and support for action to protect the environment use ordinal scales and include 'don't know' categories. A multinomial regression is used to estimate the effects of populist attitudes for these two variables. Although there is some logical order, 'don't know' should not be left out as it contains relevant information. For the question on the perceived trade-off between environmental protection and economic growth, I use an ordinary least-squares (OLS) regression to estimate the correlation between populist attitudes and this dependent variable. All models use the relevant survey weights.[3] Table A3 in the Appendix provides a bivariate correlation matrix for all metric variables.

Empirical evidence

Here, I present empirical evidence on the association between populism and stances on climate skepticism and environmental protection, first the results for climate skepticism followed by those relating to support for environmental protection.

Climate skepticism

Figure 1 shows that individuals who exhibit strong populist attitudes are systematically less likely to believe that climate change is caused by human activity compared to individuals who exhibit weak populist attitudes (top-left panel of Figure 1). In return, these individuals are more likely to believe that climate change exists but is not human-induced (top-right panel of Figure 1) and that the climate is not changing (bottom-left panel of Figure 1). The respective regression results (Table 3) confirm this.[4] Finally, individuals who strongly exhibit populist attitudes are not more likely to respond 'don't know' (bottom-right panel in Figure 1 and Table 3).

To summarise, individuals who strongly exhibit populist attitudes are more likely to be skeptical about climate change, which lends support to Hypothesis 1a.

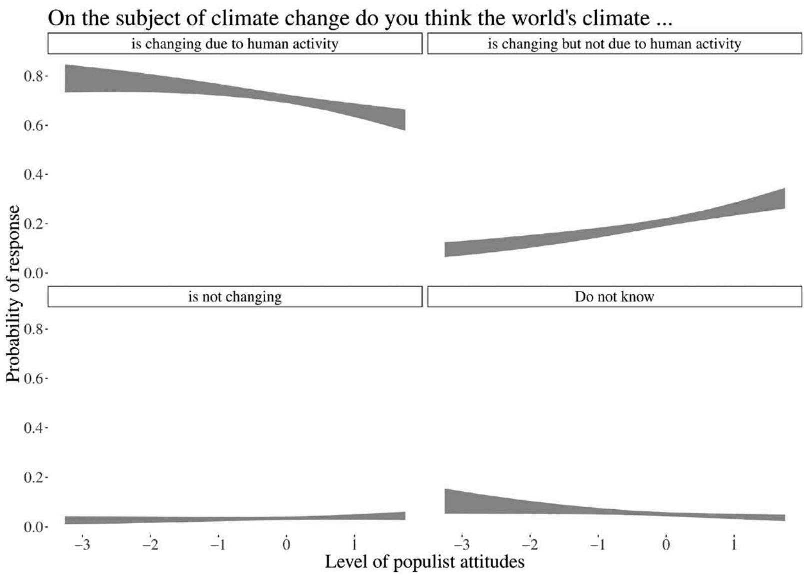

Figure 1. Populism and climate skepticism.

Each ribbon represents the 95% confidence intervals.

Table 3. Populism and climate skepticism.

	Climate is changing but not does to human activity	Climate is not changing	Don't know
Populism	0.29 (0.05)***	0.18 (0.11)*	−0.15 (0.09)
Political Ideology	0.18 (0.03)***	0.20 (0.05)***	0.09 (0.05)*
Num. obs.	3766	3766	3766

*** p < 0.01, ** p < 0.05, * p < 0.1. Entries are unstandardised coefficients from a multinomial regression. Standard errors in parenthesis. For the full regression tables including all covariates, see Table A4 in the Appendix.

Support for environmental protection

For support for environmental protection, a similar pattern emerges. Individuals who strongly exhibit populist attitudes are systematically more likely to state that measures to protect the environment have gone too far, compared to individuals who weakly exhibit populist attitudes (see top-left panel of Figure 2 and Table 4). At the same time, these populist individuals are systematically less likely to argue that government has not done enough (see bottom-left panel of Figure 2 and Table 4).

The third dependent variable measures support for environmental protection vis-à-vis economic growth. Table 4 shows a positive and statistically significant (10% level) coefficient. In other words, individuals who strongly exhibit populist attitudes are more likely to prefer economic growth, although the substantial effect size is small (also see Figure A1).

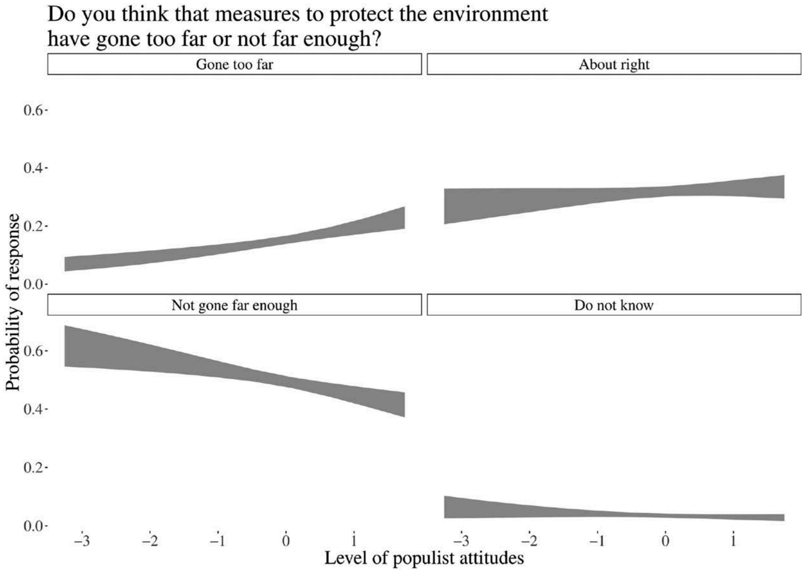

Figure 2. Populism and support for environmental policy.
Each ribbon represents the 95% confidence intervals.

Table 4. Populism and support for environmental policy.

	Do you think that measures to protect the environment have … ?			
	Gone too far	Gone not far enough	Don't know	Environmental Protection vs. Economic Growth
Populism	0.20 (0.06)***	−0.13 (0.05)**	−0.19 (0.12)	0.10 (0.05)*
Political Ideology	0.12 (0.03)***	−0.17 (0.03)***	−0.02 (0.06)	0.19 (0.03)***
Num. obs.	3766	3766	3766	3681

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Entries are unstandardised coefficients from a multinomial regression (for the first three columns) and from an OLS regression (last column). Standard errors in parenthesis. For the full regression tables including all covariates, see Table A5 in the Appendix.

In sum, populism is negatively associated with support for environmental protection, which supports Hypothesis 1b. Individuals from the UK who strongly exhibit populist attitudes tend to claim that action to protect the environment has gone too far already and tend to prioritise economic growth over environmental protection.

The role of the host-ideology

After establishing the unconditional effect of populist attitudes on climate skepticism and support for environmental protection, the question of whether this effect is conditional on political ideology arises. Lockwood (2018) explicitly argues that right-wing populists are more climate skeptic, aligning with previous research suggesting that right-wing individuals, in general, tend to be more skeptical about climate change and environmental protection (Neumayer 2004, McCright *et al.* 2016). However, understanding whether populism is similarly associated with climate and environmental attitudes across different political worldviews is important. If the data suggested the correlation on only one end of the political spectrum, this would undermine my argument and suggest that populism is *not* orthogonal but rather an enhancer of political ideology effects. To account for potential moderation effects and to better understand this relationship, an interaction term of populist attitudes and political ideology is included.

Figure 3 (also see Table A6) shows how the association of populism (light and dark grey ribbon) differs throughout a range of political ideology. For both, stating that the climate is changing due to human activity (top-left panel of Figure 3) and stating that the climate is changing but not due to human activity (top-right panel of Figure 3), I find noticeable differences between individuals who strongly (dark grey) and weakly (light grey) exhibit populist attitudes. However, by and large, there is no significant interaction between political ideology and populism. The findings for climate denial (bottom-left panel of Figure 3) and 'don't know' responses (bottom-right panel of Figure 3) are similar. This evidence suggests that populism is an important explanation of climate skepticism for left- *and* right-wing individuals.

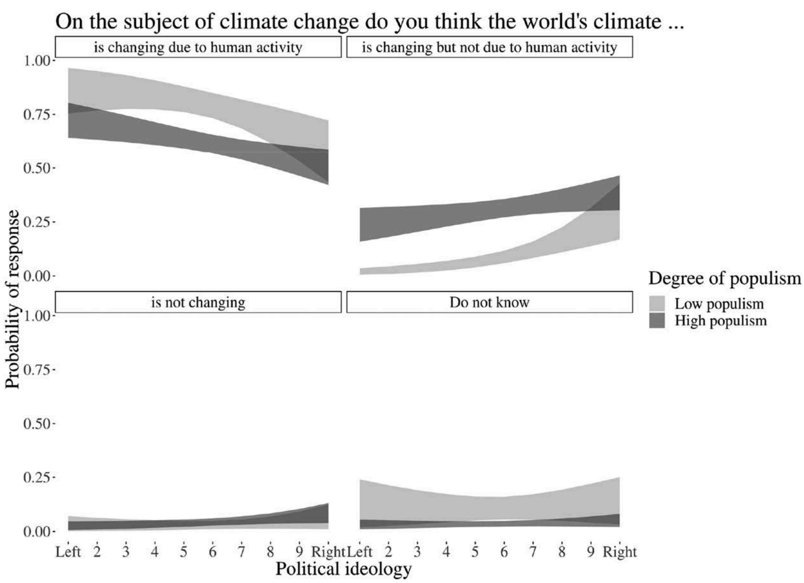

Figure 3. Populism, political ideology and climate skepticism.

Each ribbon represents the 95% confidence intervals. Colours indicate low (minimum, light grey) and high (maximum, dark grey) levels of populist attitudes.

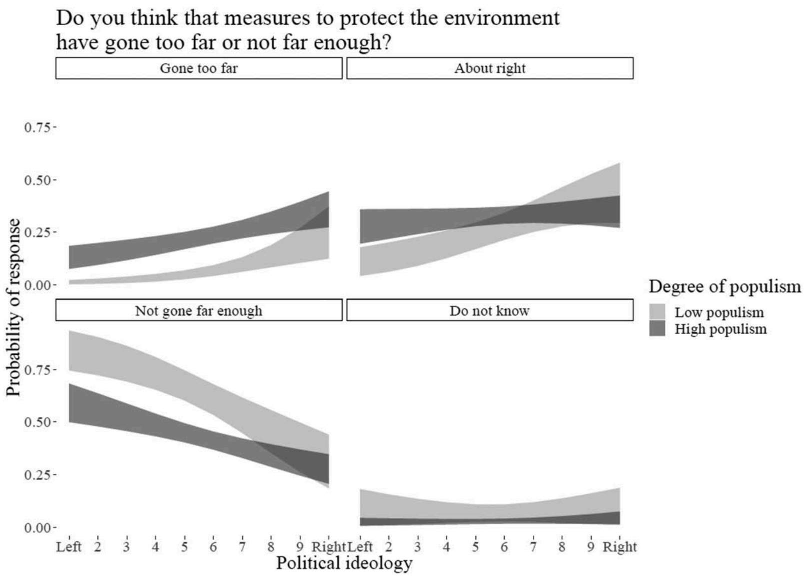

Figure 4. Populism, political ideology and support for environmental protection.
Each ribbon represents the 95% confidence intervals. Colours indicate low (minimum, light grey) and high (maximum, dark grey) levels of populist attitudes.

I observe a similar pattern when looking at environmental protection. The gap between populist and non-populist individuals is constant for left-wing and right-wing UK citizens (see Figure 4, also see Table A7 in the Appendix). Looking at the bottom left panels in Figure 4, right-wing individuals are generally less likely to agree that action has not gone far enough. However, no clear interaction between populism and political ideology is visible. The pattern for the bottom right panel and 'Don't know' responses is similar.

Finally, Figure 5 summarises the interaction of populism and host-ideology and its correlation with support for environmental protection vis-à-vis economic growth. I find no evidence of a conditional relationship between populist attitudes and political ideology.

I have shown that populist attitudes correlate with attitudes towards climate and environmental politics. To some extent, this article challenges the prevailing focus on political ideology. To get a better sense of each variable's predictive power, I assess the changes in predicted probabilities when an individual would move from a variable's minimum to maximum, all other variables held at the mean. Figures A6 and A7 (see Appendix) show that while populism is not performing as well as political ideology, its predictive power is substantial. This provides further evidence of the importance of populism in climate and environmental attitudes.

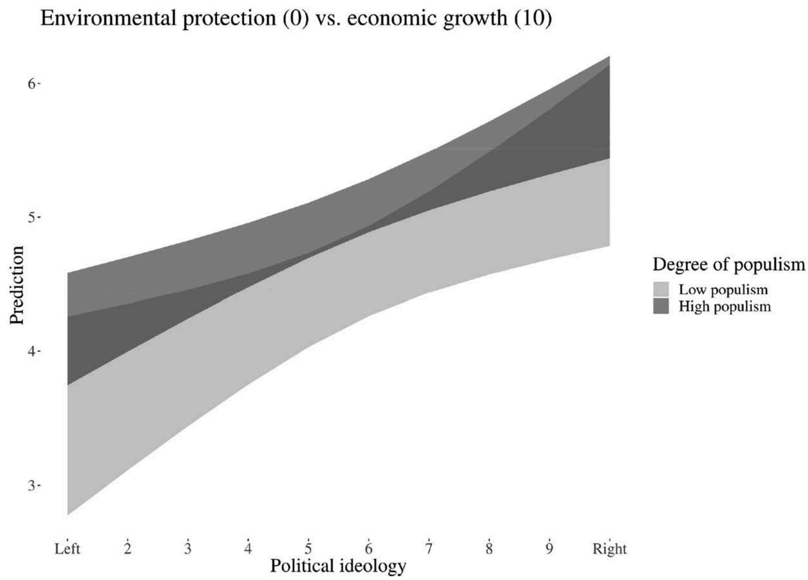

Figure 5. Populism, political ideology and environmental protection vis-á-vis economic growth.

Each ribbon represents the 95% confidence intervals. Colours indicate low (minimum, light grey) and high (maximum, dark grey) levels of populist attitudes.

Robustness checks

In order to test the robustness of the findings to alternative model specifications, five additional tests were constructed. First, I utilised a different measure for the political left–right orientation. The BES includes an alternative measure based on five questions regarding income redistribution, big business, and the rule of law. The additive score creates a new left–right scale. This test allows to specifically test whether the results are sensitive to the standard self-identified political ideology measure, which is skewed to the right, compared to this alternative measure which is skewed to the left (also see Figure A2). Second, I reran the analyses without the extreme values (1 and 10) on the ten-point left–right self-placement scale. These extreme values could potentially drive the effects of populism. Third, I reran the models for climate skepticism and support for environmental protection using an ordinal regression estimation (and logit for climate skepticism). Dropping the 'don't know' option leads to an ordinal variable. Compared to the multinomial regressions outlined earlier, this specification accounts the ordinal nature of the measures of climate skepticism and the support for action to protect the environment. Fourth, as the main independent variable stems from an estimation (in this case a factor analysis), I explicitly model the uncertainty of this estimation via bootstrapping. The initial regression analysis was re-estimated 1,000 times, with newly drawn samples (with replacement) and a new factor analysis each time. This enables a better understanding of the coefficient's sensitivity to the specificity of the sample and the uncertainty of the factor analysis. Finally, and related to the previous point, I modelled the analysis in a simple structural equation model, which confirms my findings (see Tables A15 and A16 in the Appendix).

Overall, the results remain stable throughout several different configurations (see the corresponding sections in the Appendix) and lend additional support to the association of populism and the interaction of populism with political ideology.

Conclusion

I have examined the association between populist attitudes, climate skepticism and support for environmental protection, arguing that, beyond the traditional explanation of political ideology, populist attitudes are associated with climate skepticism and stances on environmental protection. Climate politics, and to a lesser degree environmental protection, provide ideal targets for populists who can frame these issue areas as elite projects (Lockwood 2018). Individuals who strongly exhibit populist attitudes perceive a lack of representation in these issue areas and thus, because of their anti-elitist stance, reject climate and environmental policies. In other words, rejecting the elite tends to be associated with climate skepticism and lower support for environmental protection.

Data from the British Election Study offer ample support for the claim that individuals who strongly exhibit populist attitudes also tend to be more skeptical concerning climate change. Populist attitudes also explain variation in support for environmental protection. These findings illustrate the importance of populism in explaining individuals' attitudes towards climate change and environmental politics. The effect of populism seems to be independent of political ideology. Altogether, it appears populist attitudes could potentially endanger public support for future climate action and new, binding international treaties beyond specific combinations of populism and political ideology.

Considering populism as an explanation for differing stances on climate change and environmental protection is of utmost importance for four reasons. First, populism and individuals' populist attitudes are theoretically independent of specific ideological left–right positions and not confounded by them (Mudde 2004, Rooduijn and Akkerman 2017, Van Hauwaert and Van Kessel 2018), which might explain why individuals could be skeptical about climate change and environmental protection beyond current explanations concerning political ideology (Lockwood 2018). Second, current explanations neglect the central actors in climate action and environmental protection. With the focus on populism, I have here directly confronted the central actor at the heart of policymaking. Third, the conceptualisation of the people and the elite differs vastly across left-wing and right-wing populism. Consequently, it is plausible that the correlation of populism is weaker for some ideologies in comparison to others. Finally, Hornsey *et al.* (2016) highlight that political factors outperform sociodemographic variables in explaining stances on climate change (Beiser-McGrath and Huber 2018). Despite the importance of populism, explanations of political ideologies have focused mainly on party identification, and the differences between conservatives and liberals, which are crude proxies for the varieties of political ideologies worth considering.

Research on populist attitudes shares the notion that these attitudes are widespread; events, such as Brexit in the United Kingdom, are regularly described as a *mutiny against the elite* (see Calhoun 2016, Hobolt 2016). Thus, further understanding how populism relates to policy questions is essential. Indeed, the top-down logic of climate policies might substantially undermine public support. Elites negotiate international treaties in the hope that individuals will perceive these signals of encouragement to adapt their behaviour; but although they seek to increase awareness of climate change, the public is largely excluded, which in turn leads to anti-elitism affecting public support for these policies in a populist backlash (as Norris and Inglehart 2018 identify for globalisation). Thus the way elites communicate the negative effects of climate change could backfire (Hart and Nisbet 2012); and in order to reach individuals who strongly exhibit populist attitudes, governments must assess different ways of making and communicating these political decisions. In this context, the greater inclusion of citizens in decision-making may help overcome current problems in garnering enough public support for climate and environmental politics (Romsdahl *et al.* 2018).

This implication ties in directly to the discussion about the 'post-truth era' (Keyes 2013). Initially, the communication of scientific evidence was expected to suffice as the necessary transmission belt to improve literacy in science. As long as communication is perceived as a tool to sell elite positions or to persuade 'ordinary citizens', it is bound to fail to reach certain groups. The 2016 EU referendum campaign in the UK and the US presidential campaign have illustrated that facts seem to be less significant in evaluating political positions; if, as Drezner (2016) argues, '[t]he marshalling of undisputed facts and evidence doesn't have quite the effect on public debate that it used to', it may follow that facts are no longer a transmission belt to transform climate skeptics into believers; and if truths are not the core foundation for building beliefs, they are also unlikely to sway public opinion. Thus, the underlying problem may not simply be solved by better presenting and communicating scientific results.

Notes

1. One limitation of this measure is that it fails to acknowledge the potential multidimensionality of climate skepticism (Capstick and Pidgeon 2014). Table A14 in the Appendix suggests that recoding this variable into those who believe in climate change and those who do not does not affect the findings.
2. This measurement neglects potential synergies between environmental protection and economic growth. This could be one reason why the explained variance for this variable is rather small ($R^2 = 0.16$).
3. The statistical package *R* was used for data manipulation and analysis (R Core Team 2015). Replication files are available under: https://doi.org/10.7910/DVN/4YU9LJ.
4. Note that for the category 'climate is not changing,' populism is significant at the 10 per cent level, but fails to reach the conventional 5 per cent level.

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